METRIC
MIL-P-22088B(SH)
15 April 1993
SUPERSEDING
MIL-P-22088A(SH)
7 March 1989
(See 6.14)

MILITARY SPECIFICATION

PURIFIERS, JET FUEL CENTRIFUGAL, NAVAL SHIPBOARD

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the construction, performance, and delivery acceptability for centrifugal jet fuel purifiers for service on board Naval surface ships.

1.2 Classification. Centrifugal purifiers are of the standard size specified in table I.

TABLE I. Purifier size.

Purifier size	Capacity 1/	Fluid
JF-3	1136 (300)	JP-5 (see MIL-T-5624)

1/ Capacity in liters per minute (L/min) (gallons per minute (gal/min)).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, SEA 05Q42, Naval Sea Systems Command, 2531 Jefferson Davis Hwy, Arlington, VA 22242-5160 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4330

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

FF-B-171	- Bearings, Ball, Annular (General Purpose).
FF-B-185	- Bearings, Roller, Cylindrical; and Bearings, Roller, Self-Aligning.
QQ-C-390	- Copper Alloy Castings (Including Cast Bar).
QQ-S-766	- Steel, Stainless and Heat Resisting, Alloys, Plate, Sheet and Strip.
WW-P-404	- Pipe, Steel, (Seamless and Welded, Black and Zinc-Coated (Galvanized)).
PPP-F-320	- Fiberboard: Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

MILITARY

MIL-S-901	- Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for.
MIL-E-917	- Electrical Power Equipment, Basic Requirements (Naval Shipboard Use).
MIL-C-2212	- Contactors and Controllers, Electric Motor AC or DC, and Associated Switching Devices.
MIL-T-5624	- Turbine Fuel, Aviation, Grades JP-4, JP-5 and JP-5/JP-8 ST.
MIL-S-12875	- Sheets, Perforated, Metal.
MIL-P-15024	- Plates, Tags and Bands for Identification of Equipment.
MIL-P-15024/5	- Plates, Identification.
MIL-P-16789	- Pumps (Including Prime Movers and Support Items); Packaging of.
MIL-M-17060	- Motors, 60-Hertz, Alternating Current, Integral-Horsepower, Shipboard Use.
MIL-I-17244	- Indicators, Temperature, Direct-Reading Bimetallic, (3 and 5 Inch Dial).
MIL-M-17508	- Mounts, Resilient: Types 6E100, 6E150, 7E450, 6E900, 6E2000, 5E3500, 6E100BB, 6E150BB, 7E450BB, 6E900BB.
MIL-B-17931	- Bearings, Ball, Annular, For Quiet Operation.
MIL-G-18997	- Gauge, Pressure, Dial-Indicating.
MIL-L-19140	- Lumber and Plywood, Fire-Retardant Treated.
MIL-P-24441	- Paint, Epoxy-Polyamide General Specification for.
MIL-P-24441/1	- Paint, Epoxy-Polyamide, Green Primer, Formula 150, Type I.

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MIL-P-24441/2 - Paint, Epoxy-Polyamide, Exterior Topcoat, Haze Gray, Formula 151, Type I.
MIL-P-24691 - Pipe and Tube, Carbon, Alloy and Stainless Steel, Seamless and Welded, General Specification for.
MIL-P-24691/3 - Pipe and Tube, Corrosion-Resistant, Stainless Steel, Seamless or Welded.
MIL-H-46855 - Human Engineering Requirements for Military Systems, Equipment and Facilities.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.
FED-STD-H28/2 - Screw-Thread Standards for Federal Services Section 2 Unified Inch Screw Threads - UN and UNR Thread Forms.

MILITARY

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-278 - Welding and Casting Standard.
MIL-STD-454 - Standard General Requirements For Electronic Equipment.
MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-470 - Maintainability Program for Systems and Equipment.
MIL-STD-471 - Maintainability Verification/Demonstration/ Evaluation.
MIL-STD-721 - Definitions of Terms for Reliability and Maintainability.
MIL-STD-777 - Schedule of Piping, Valves, Fittings, and Associated Piping Components for Naval Surface Ships.
MIL-STD-781 - Reliability Testing for Engineering Development, Qualification, and Production.
MIL-STD-785 - Reliability Program for Systems and Equipment Development and Production.
MIL-STD-1472 - Human Engineering Design Criteria for Military Systems, Equipment, and Facilities.

HANDBOOKS

MILITARY

MIL-HDBK-267 - Guide for Selection of Lubricants and Hydraulic Fluids for Use in Shipboard Equipment.

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(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, BLDG. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (NAVSEA)
NAVSHIPS 803-1385850 - Piping, Instrument Pressure for all Service.
NAVSHIPS 810-1385917 - Thermometer Selection Guide.

(Application for copies should be addressed to: Commander, Portsmouth Naval Shipyard, NEDSA, Code 202.2, Portsmouth, NH 03804-5000.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 36 - Standard Specification for Structural Steel. (DoD adopted)
A 106 - Standard Specification for Seamless Carbon Steel Pipe for High-Temperature Service. (DoD adopted)
A 108 - Standard Specification for Steel Bars, Carbon, Cold Finished, Standard Quality. (DoD adopted)
A 249 - Standard Specification for Welded Austenitic Steel Boiler, Superheater, Heat-Exchanger, and Condenser Tubes.
A 322 - Standard Specification for Steel Bars, Alloy, Standard Grades. (DoD adopted)
A 473 - Standard Specification for Stainless and Heat-Resisting Steel Forgings. (DoD adopted)
A 568 - Standard Specification for Steel, Sheet, Carbon, and High-Strength, Low-Alloy, Hot-Rolled and Cold-Rolled, General Requirements For. (DoD adopted)
A 569 - Standard Specification for Steel, Carbon (0.15 Maximum, Percent), Hot-Rolled Sheet and Strip Commercial Quality. (DoD adopted)
D 2276 - Standard Test Method for Particulate Contaminant in Aviation Turbine Fuels. (DoD adopted)
D 2709 - Standard Test Method for Water and Sediment in Distillate Fuels by Centrifuge. (DoD adopted)

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ASTM (Continued)
D 3240 - Standard Test Method for Undissolved Water in Aviation Turbine Fuels. (DoD adopted)
D 3370 - Standard Practices for Sampling Water. (DoD adopted)
D 3921 - Standard Test Method for Oil and Grease and Petroleum Hydrocarbons in Water.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)
J726 - Recommended Practice for Air Cleaner Test Code. (DoD adopted)

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

STEEL STRUCTURES PAINTING COUNCIL (SSPC)
SP 10 - Near-White Blast Cleaning.

(Application for copies should be addressed to Steel Structures Painting Council, 4400 Fifth Avenue, Pittsburgh, PA 15213.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.5) in accordance with 4.3.

3.2 Materials. Purifier materials shall be as specified in table II. However, this specification is not intended to be restrictive, provided proposed alternate materials are equal to or better than the materials specified. Proposed alternate materials shall be subject to review and acceptance by the contracting activity. Parts for which specific materials are not specified shall be made of materials similar to those cited for the service intended. Use of dissimilar materials shall be avoided.

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TABLE II. Purifier materials.

Application	Material	Specification
Frame	Hot rolled steel	ASTM A 569 WW-P-404, class A 53AS ASTM A 36
	Hot rolled and cold rolled steel	ASTM A 568
	Cold rolled steel	ASTM A 108
Covers	Bronze, hydraulic	QQ-C-390
	Steel	ASTM A 108, grade 1018 ASTM A 569 MIL-S-12875, type VII ASTM A 36
	Stainless steel	QQ-S-766, class 302, 304
Bowl assembly	Stainless steel	QQ-S-766, class 316 AISI 329
Bowl casing	Hot rolled steel	ASTM A 473, ASTM A 36, ASTM A 569
	Hot rolled and cold rolled steel	ASTM A 568
	Steel pipe	WW-P-404, class A 53AS ASTM A 106, grade B
	Cold rolled steel	ASTM A 108, grade 1018
Spindle	Steel	ASTM A 108 ASTM A 322, grade E43406
Inlet and outlet piping	Stainless steel 304	MIL-P-24691 and MIL-P-24691/3, ASTM A 249 grade TP 304

3.2.1 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been

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collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.2.2 Prohibited material. The following materials shall be prohibited for use in purifiers covered by this specification:

(a) Magnesium and its alloys.
(b) Cast iron (except nodular iron).
(c) Semi-steel.
(d) Cadium plating.
(e) Asbestos.

3.2.3 Mercury. The purifier shall be free of mercury contamination. During the manufacturing process, tests, and examination the product to be offered for acceptance shall not come in direct contact with mercury or any of its components, nor with any mercury containing device employing a single boundary of containment (see 6.3).

3.2.4 Ancillary components. Materials for ancillary components, such as motors and controllers, shall be in accordance with the specification for those components.

3.3 Construction. Purifiers shall be constructed to separate contaminants and water from jet fuel and discharge purified fuel under pressure through closed piping to the ship's fuel system (see 6.3 and 6.4). Water separated from the fuel shall be continuously discharged to the ship's oily water drain collection system. Separated solids shall collect within the bowl until they are removed by manual cleaning. The purifier shall be constructed for simplicity of operation and repair and shall perform in accordance with design characteristics (see 3.5.1) under all shipboard conditions (see 3.6).

3.3.1 General. Purifiers shall have the following characteristics:

(a) Manual push-button motor start (control panel characteristic).

(b) Electric motor drive through gears.
(c) Speed indicator or counter for measurement of bowl speed.

3.3.2 Mounts. Unless otherwise specified (see 6.2), purifiers shall be vertical and hard-mounted on a base plate and provided complete, ready for mounting directly to the deck of the ship without the use of a pedestal.

3.3.2.1 Resilient mounts. When specified (see 6.2), resilient mounts shall be in accordance with MIL-M-17508.

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3.3.3 Frames. Purifier frames shall be sufficiently rigid to allow lifting, handling, shipment, and installation of the units on board ship without disturbing the alignment of the assembled units such that the normal distortion, weaving, or vibration of the supporting structures on board ships cannot cause misalignment of rotating elements.

3.3.3.1 Special tools or fittings. Special tools or fittings shall be provided by the manufacturer for lifting all items weighing in excess of 16 kilograms (kg) (35 pounds).

3.3.4 Brakes. Purifiers shall be equipped with bowl brakes.

3.3.5 Bearings. Bearing installation and lubrication shall be in accordance with FF-B-171 or FF-B-185. For low noise applications, bearings shall be in accordance with the construction requirements of MIL-B-17931. Bearing assemblies shall meet the vibration requirements for all bearings in accordance with MIL-B-17931.

3.3.5.1 Bearing types. Purifier bowl spindle bearings shall be ball bearings. Other bearings may be of the sliding contact or rolling contact types. Double row bearings and bearing sets shall be prohibited unless they are the self-aligning type.

3.3.5.2 Bearing selection. In the selection of bearings, consideration shall be given to the fact that rolling and pitching of a ship at sea may introduce unusual radial and axial thrust loads.

3.3.6 Lubricants. Lubricants used for purifier operation, assembly, and maintenance shall be selected from those standard lubricants listed in MIL-HDBK-267.

3.3.7 Piping and valves. Connections to ship's piping shall be flanged. Purifier valves, flanges, piping, bolting gaskets, and fittings interfacing with ship's systems shall be in accordance with MIL-STD-777.

3.3.8 Prime water connection. A connection shall be provided on purifiers for purifier bowl prime. The quantity for prime water shall be designated by the manufacturer. The construction shall minimize water requirements.

3.3.9 Fuel/water interface control. Control of the interface between liquid seal (prime water) and fuel shall be achieved by means of a backpressure regulating valve and gravity discs. Under all operating conditions back pressure shall be sufficient to prevent aeration and foaming of purified oil.

3.3.10 Adjustment for specific gravity of fuel. The purifier shall include a set of gravity discs for specific gravity adjustment to ensure achieving the highest possible degree of purification and clarification for any specific fuel condition.

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3.3.10.1 Adjustment range. The range of adjustment shall cover a gravity range of from 34 degrees American Petroleum Institute (API) (0.855 specific gravity) to 50 degrees API (0.780 specific gravity). Any graph or table provided shall be based on API degrees and specific gravity (see 6.4).

3.3.11 Drain connection. A connection shall be provided for a casing drain.

3.3.12 Observation port. An observation port shall be provided in the heavy phase discharge.

3.3.13 Pressure gauges, gauge piping and connections. Pressure gauges shall be 9 centimeters (cm) (3.5 inches) in diameter and shall be in accordance with MIL-G-18997. Gauge piping and connections shall be in accordance with Drawing 803-1385850. Gauges shall be installed to resist breakage and disconnection caused by damaging vibration.

3.3.14 Thermometers and thermometer wells. Thermometer wells and bulbs, when supplied, shall be in accordance with Drawing 810-1385917.

3.3.14.1 Thermometers. Thermometers shall be 7.6 cm (3 inches) in diameter in accordance with MIL-I-17244. Use of mercury thermometers shall be prohibited.

3.3.15 Oil dipstick and sight glass. Purifiers shall be provided with an oil dipstick or sight glass to determine the lube oil level within the drive mechanism case.

3.3.16 Thread fit. Screw threads shall conform to FED-STD-H28 and FED-STD-H28/2.

3.3.17 Doweling. Each component part of an assembled unit supported directly by a frame or bedplate shall be doweled to facilitate reassembly and maintenance of alignment. Dowels shall be unequally spaced or of different diameters to avoid incorrect assembly.

3.3.18 Machined surfaces. Bearing and seating surfaces of frames or bedplates shall be finish machined.

3.3.19 Welding. Welding shall be in accordance with MIL-STD-278. Loose welding beads, slag, and spatter shall be removed.

3.3.20 Painting. Non-working surfaces, normally painted, shall be finished as specified in the following procedures:

(a) Surface preparation prior to painting in accordance with SSPC SP 10.
(b) Two coats of epoxy paint formula 151 and one coat of formula 150 in accordance with MIL-P-24441, MIL-P-24441/1, and MIL-P-24441/2. Average 3 mils dry film thickness per coat.

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3.3.21 Motors. Motors shall be in accordance with MIL-M-17060 for alternating current (ac) motors and shall be of the following characteristics:

Horsepower	See 3.3.21.1.
Service	A.
Speed classification	Constant.
Duty	Continuous.
Enclosure	Totally enclosed or dripproof.
Bearings	In accordance with FF-B-171; MIL-B-17931 if low noise application.
Ambient temperature	50 °C (122°F).
Insulation	B or F, with sealed insulation system, or class H.
Type	Squirrel cage induction, wye-delta (see 6.2).
Voltage rating	Unless otherwise specified (see 6.2), 440 volts ac, 3-phase.
Design	As required (see 3.3.23).
Low noise insulation	As specified (see 6.2).

3.3.21.1 Horsepower rating. The horsepower rating of each motor shall be not less than the sum of the maximum brake horsepower of the driven components under any condition of service. The actual motor rating shall be in accordance with MIL-M-17060. If the maximum brake horsepower of the equipment is less than 2, and if there is a possibility of an unpredictably high frictional load due to an improper adjustment of glands or to some other cause, then the maximum brake horsepower shall be multiplied by a safety factor of 1.5-maximum brake horsepower/4 before selecting the next larger Navy standard rating.

3.3.21.2 Thermal protection. Thermal protection shall be provided for the purifier motor stator and motor bearings in accordance with MIL-M-17060, with the provision for de-energizing the motor control. Thermal protection for motor bearings and motor stator shall be in parallel electrically. The thermal protection shall include motor-mounted monitor control with a plug-in relay having not less than one pair of normally open and one pair of normally closed contacts. The temperature sensing system shall include not less than three thermal sensors imbedded in the stator windings and one thermal sensor for each motor bearing. Positive temperature coefficient (ptc) thermal sensors shall be used and shall have a temperature coefficient of resistance at the switchpoint of not less than 15 percent per degree Celsius.

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3.3.22 Controllers. Controllers shall be in accordance with MIL-C-2212 and shall be of the following characteristics:

Enclosure	Dripproof.
Operation	Magnetic.
Type	Wye-delta.
Function	Motor starting.
Duty	Continuous.
Protection	Overload - under voltage (LVP). No overload protection during start period.
Ambient temperature	50°C (122°F).
Voltage	Unless otherwise specified (see 6.2), 440 volts ac, 3-phase; direct current (dc) voltage requirements, if applicable, shall be as specified (see 6.2).
Performance	Manual.
Master switches	Local, manual.
Abnormal vibration	When not covered by MIL-C-2212, frequency shall be as specified (see 6.2).
Low noise application	When specified (see 6.2), insulation shall be class B with class A for coils only.

3.3.23 Acceleration allowance. Motor and controller construction shall permit acceleration of the unloaded purifier to full operating speed in less than 12 minutes, without exceeding the permissible temperature rise in any components or causing wear due to slippage of drive. Overload relays shall be effective for all periods other than the starting period. They shall be the compensated thermal type.

3.3.24 Interchangeability. All identically identified components shall be functionally and physically interchangeable, without degradation of performance, reliability, or operating characteristics and without selective assembly or modification except for calibration and adjustment. Repair parts shall be interchangeable with, and identified identically to, the part they replace (see 6.3). Notwithstanding the above, parts balanced as complete assemblies shall not be exchanged without rebalancing the complete assembly.

3.3.25 Accessibility. The construction shall provide the maximum accessibility (within space limitations) to parts which require scheduled maintenance and to effect repairs. Construction shall be such that the purifier can be opened for cleaning and maintenance without breaking any piping connections. Removal of major operational parts shall not be necessary to permit accessibility for maintenance, inspection, and repair. The human engineering considerations specified in MIL-STD-1472 and MIL-H-46855 shall not be degraded during detail design and construction.

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3.4 Dimensions. Purifiers shall not exceed the dimensions of table III, including space for disassembly of parts but exclusive of operating space.

TABLE III. Limiting dimension.

Dimension	Meters (feet)
Length	1.68 (5.5)
Width	1.22 (4.0)
Height	2.13 (7.0)

3.5 Performance.

3.5.1. Rated capacity. Unless otherwise specified (see 6.2), the purifier shall purify fuel at it's rated capacity, as specified in table I.

3.5.1.1 Surplus capacity. The purifier shall purify fuel in excess of at least 15 percent of it's rated capacity. (This requirement is necessary to accommodate transfer pump flow rates that exceed their rated capacity and provides the design activity a limited degree of flexibility when sizing the fuel transfer pump).

3.5.2 Emergency capacity. The purifier shall be constructed so that it handles 200 gallons per minute (gal/min) water without danger of any water being discharged out of the clean fuel outlet. The purifier shall discharge 200 gal/min of water continuously through the heavy phase discharge.

3.5.3 Solids retention. The total solids retention volume of the purifier shall be not less than 15 liters (L) (4 gal).

3.5.4 Shock. The purifier shall be constructed to preclude failure of parts or appendages when subjected to test requirements in accordance with MIL-S-901 for grade A, class II, type A. As a minimum, the purifier shall restart and resume normal operation if it shuts down when subjected to test requirements in accordance with MIL-S-901 for grade A, class II, type A. Equipment shall be hard-mounted to a base for test purposes. The determination of stresses resulting from the shock test shall be accomplished by the application of analytical mechanics. The allowable stress in components due to shock shall be the yield strength (0.2 percent offset of the material). Normal stresses in parts, such as the tensile stress in a bolt due to initial pull-up, shall not be

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added to the shock stresses. The maximum accelerations possible under the shock tests shall be utilized in the construction of hold-down bolts, supporting feet or lugs, and main structural member of the equipment. No failure or permanent deformation shall occur as a result of shock. Unless the manufacturer has reason to believe that higher accelerations may occur, components of an assembly mounted on a common rigid base need not be constructed for accelerations higher than the values for which the whole assembly is constructed.

3.5.5 Mechanical vibration. Equipment shall be free of self-induced destructive vibration under all operating conditions. The equipment shall be in accordance with type II of MIL-STD-167-1.

3.5.5.1 Environmental vibration. The equipment shall withstand the environmental vibration requirements of MIL-STD-167-1, type I (see 6.2).

3.5.6 Electromagnetic interference. The purifier shall conform to the electromagnetic interference suppression requirements of MIL-STD-461 for surface ships (class A4).

3.6 Environmental conditions. Equipment shall operate continuously at rated capacity under any combination of the range of environmental conditions specified herein.

3.6.1 Temperatures. Temperatures shall be as specified in table IV.

TABLE IV. Temperatures.

	JP-5 fuel °C (°F)
Influent temperature range	15-32 (60-90)
Ambient temperature range	4-38 (40-100)

3.6.2 Pitch and inclined operation. The purifier shall suffer no damage and shall operate according to the requirements specified herein when subjected to the motions specified in table V.

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TABLE V. Motions 1/

Motion attitude	Radians (degrees)
Pitch (from normal either direction)2/	0.17 (10)
Trimmed down (from normal horizontal)	0.087 (5)
Roll (from vertical)3/	4/ 0.785 (45)
Permanently listed (either side of vertical)	0.262 (15)

1/ Trim and list conditions shall be considered to occur simultaneously. Roll and pitch conditions shall be considered to occur simultaneously.
2/ Time of cycle, 10 seconds.
3/ Time of cycle, 15 seconds.
4/ The purifier shall maintain lubrication, without loss of oil to the drive assembly. The purifier shall continue to operate, without the need for repriming, after recovering from the subject attitude.

3.7 Reliability and maintainability. Definitions of terms for reliability and maintainability shall be in accordance with MIL-STD-721, MIL-STD-785, and MIL-STD-470.

3.7.1 Reliability. Mean time between failure (MTBF) for the purifier shall be not less than 1500 hours (see 6.3).

3.7.2 Maintainability. Mean time to repair (MTTR) for the purifier shall be not greater than 2 hours (see 6.3).

3.7.3 Failure. For purposes of reliability calculation, the purifier shall be considered to fail if it cannot be started, or if a malfunction occurs that forces the purifier to be shut down prior to the completion of its scheduled test time or operating time. In addition, the purifier shall be considered to fail if the capacity (see 3.5.1) or the quality of purification (see 4.6.8.2) is not in accordance with the requirements specified herein, even if the unit has not experienced a forced shutdown. Failures shall be classified as either relevant or nonrelevant. Only relevant failures shall be chargeable for verifying reliability requirements.

(a) Relevant failures are caused by:

(1) Equipment design defects.
(2) Equipment manufacturing defects.
(3) Parts defects.
(4) Unknown.
(5) All other causes not specifically listed as non-relevant.

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(b) Non-relevant failures are caused by:

(1) Accident, mishandling, or improper storage.
(2) Operator or procedural error.
(3) External test equipment or facility failures.
(4) Drawing, maintenance manual, or other documentation errors, provided prompt correction is shown to eliminate future similar failures (this shall not be construed to include design errors).
(5) Failures of multiple, simultaneous or immediately sequential nature (only the initial failure shall be counted in such cases).
(6) Direct result of relevant or non-relevant failures of another item of equipment (secondary failures).
(7) Improper maintenance-induced failures.
(8) Foreign object damage, unless caused by failure of protective device built into contractor's equipment.

Failure reporting shall include all failures, whether non-relevant or relevant.

3.8 Identification plates. Identification plates shall be provided on each purifier and shall be type A or B in accordance with MIL-P-15024 and MIL-P-15024/5, except that plastic identification plates shall not be acceptable.

3.8.1 Securing identification plates. Identification plates shall be secured to equipment with corrosion-resistant metallic screws.

3.8.2 Identification data. Purifier identification plates shall contain the following data:

(a) Manufacturer's name.
(b) "Centrifugal jet fuel purifier" (followed by flow rate).
(c) Manufacturer's model.
(d) Navy size designation.
(e) Date of manufacture (month and year).
(f) Manufacturer's serial number.
(g) Contract number (and item number for multiple unit orders).
(h) National stock number.
(i) Nominal bowl speed.

3.8.3 Other identification plates. Each driving unit and each accessory unit shall have an identification plate in accordance with the applicable equipment specification.

3.9 Safety criteria. Equipment shall be constructed to ensure safety to operating and maintenance personnel (see 6.3). Purifiers shall have their bowls enclosed to prevent free vapor from escaping. External moving parts, which are a potential hazard to personnel, shall be avoided wherever practicable. When their use is unavoidable, positive protection in the form of a guard shall be provided. Sharp corners and projections which may cause injury to personnel or on which clothing may catch shall be avoided. The following are some of the known safety criteria that apply to auxiliary machinery; others are contained in documents such as MIL-STD-454 and MIL-E-917.

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(a) Exposed rotating or reciprocating components, such as couplings, linkages, belts, chains, and flywheels shall be enclosed with a safety guard to prevent accidental personnel contact with the moving part.
(b) When an equipment specification requires the presence of a hazardous material as a component of shipboard equipment, conspicuous indication on the equipment of the presence of such material together with necessary warnings and instructions shall be required.

3.10 Special tools. When specified (see 6.2), special tools shall be provided in the quantity specified. Special tools are defined as those items required for shipboard maintenance, servicing, and repair not listed in the Federal Supply Catalog. (Copies of this catalog may be consulted in the office of the Defense Contract Management Area Operations (DCMAO)).

3.11 Workmanship. Workmanship shall conform to requirements for specified engineering and production processes. The purifier shall withstand test operation specified herein without permanent deformation, breaks, malfunction, and component interference. Each part of the purifier, before and after painting, shall be clean and free of sand, rust, dirt, pits, sprues, scale, and other harmful extraneous material which might detract from the intended operation, function, or appearance of the equipment. Fasteners shall be fully engaged and bolt heads and nuts shall be flat on washers or mating surface.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program (see 6.3). The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

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4.3 First article inspection. First article inspection shall be performed at a laboratory acceptable to the Government on a sample purifier produced with equipment and procedures normally used in production.

4.3.1 Sampling for first article inspection. One sample unit shall be subjected to first article inspection.

4.3.2 Inspection routine. The sample unit shall be subjected to the examinations and tests specified in 4.5 through 4.6.2 and 4.6.4 through 4.6.14.

4.3.3 Failure. Failure to pass the examination (see 4.5) or failures other than those allowed during testing (see 3.7.3) shall be cause for refusal of acceptance.

4.4 Quality conformance inspection. All purifiers shall be examined and tested as specified in 4.5 through 4.6.3. The operational test (see 4.6.2) and the driver test (see 4.6.3) shall be performed concurrently.

4.5 Examinations. Each purifier shall be examined to ascertain that the equipment is in accordance with all the requirements of this specification not involving tests and as specified in 4.5.1 through 4.5.4.

4.5.1 Fabrication. Welding shall be examined to ensure conformance to 3.3.19.

4.5.2 Painting. Each purifier shall be examined at each step of the painting to ensure conformance to the requirements specified in 3.3.20.

4.5.3 Dimensions. Each purifier shall be examined to ensure conformance to the dimensions specified in 3.4.

4.5.4 Identification plate. Each purifier shall be examined to ensure conformance to the identification requirements specified in 3.8.2.

4.5.5 Workmanship. Each purifier shall be examined to ensure conformance to the requirements for workmanship specified in 3.11.

4.6 Tests. Test procedures shall be as specified in 4.6.1 through 4.6.14 (see 6.3).

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4.6.1 Spin-test. Prior to installation in the purifier, the bowl assembly shall be spin-tested at 125 percent of the maximum designed operating speed for a period of not less than 15 minutes.

4.6.2 Operational test. Each completely assembled purifier shall be operated for a period of not less than 10 minutes at rated speed to determine smoothness of running, satisfactory operation of controls, and proper assembly.

4.6.3 Driver test. Purifiers shall be operated at full capacity for a period of not less than 1 hour. Quantitative measurements of purification are not required. During this test, the brake horsepower input to the assembled purifier unit shall be determined. This test shall be performed using 100 percent water at the rated capacity.

4.6.4 Capacity test. The purifier shall be operated using grade JP-5 in accordance with MIL-T-5624, having a specific gravity of from 0.788 to 0.845 at 15°C (60°F), for not less than 1 hour to verify purifier rated capacity. Test conditions, acceptable performance, and test methods shall be as specified in 4.6.8.

4.6.4.1 Surplus capacity test. The purifier shall be operated using grade JP-5 in accordance with MIL-T-5624, having a specific gravity of from 0.788 to 0.845 at 15°C (60°F). The feed rate shall be at least 15 percent in excess of the rated capacity (that is, 345 gpm for size JF-3 purifier) for a period of not less than 1 hour. Test conditions, acceptable performance, and test methods shall be as specified in 4.6.8.

4.6.5 Reliability test. The sample unit chosen for first article testing shall be operationally tested for a total of not less than 1800 hours in accordance with the requirements specified in 3.7 (see 6.3). The test shall be run in accordance with task 202 of MIL-STD-781. Each operational period shall be not less than 7 hours. The motor controller used during this test shall be identical to the one to be supplied with the purifier. The purifier shall be operated using grade JP-5 in accordance with MIL-T-5624, having a specific gravity of from 0.788 to 0.845 at 15°C (60°F). The fuel feed rate to the purifier shall be as specified in 4.6.4. Test conditions, acceptable performance, and test methods shall be as specified in 4.6.8.

4.6.6 Maintainability test. The contractor shall demonstrate that the equipment design meets the quantitative requirements of MIL-STD-471, test method 4 (see 6.3).

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4.6.7 Inclination test. For 24 hours of the reliability test, the purifier shall be at an inclination of 0.262 radians (15 degrees).

4.6.8 Effluent quality test. Test conditions, acceptable performance, and methods of determining acceptable performance shall be as specified in 4.6.8.1, 4.6.8.2, and 4.6.8.3, respectively. An effluent quality test shall be performed on the sample unit at 10, 600, 1200, and 1800 hours of operation. The test fluid shall be as specified in 4.6.5.

4.6.8.1 Test conditions. Test conditions shall be as follows:

(a)	Flow	- As specified in 3.5.1 (plus 10 percent, minus 0 percent).
(b)	Water in oil	- 1 ± 0.1 percent. The water contaminant shall be tap water filtered through any commercial standard filter.
(c)	Solids	- AC fine dust in accordance with SAE J726, 100-200 milligrams (mg) per liter.
(d)	Temperature	- 24 ± 8°C (75 ± 15°F).

4.6.8.2 Acceptable performance, fuel quality (effluent). Alternate or modified test procedures (that is, sampling methods and analysis) shall be subject to review and acceptance by the contracting activity.

4.6.8.2.1 Solids contaminants. The purified oil shall have either (1) an average weight of solids not greater than 0.70 mg per liter, with no single sample greater than 3.0 mg per liter, or (2) a minimum solids removal efficiency of not less than 99.5 percent.

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4.6.8.2.2 Undissolved (free) water content. The average free water content in the jet fuel shall be not greater than 5.0 parts per million (p/m) for all samples.

4.6.8.3 Test methods. The influent and effluent samples shall be taken simultaneously.

4.6.8.3.1 Influent test methods. Influent quality test methods shall be performed to verify conformance to 4.6.8.1 (b) and (c) as follows:

(a) Sampling procedures. The sample cock shall be fully opened and 200 milliliters (mL) shall be drawn. The 200 mL shall be discarded. Samples shall then be obtained as follows:

(1) Undissolved (free) water (BS&W) - At least six influent samples shall be drawn directly into a 100 mL graduated centrifuge tube for free water determination.
(2) Gravimetric analysis - At least six influent samples shall be drawn into a 32 ounce bottle for gravimetric analysis.

(b) Undissolved (free) water determination (BS&W) - Undissolved water shall be determined in accordance with ASTM D 2709 (that is, centrifuge, 100 mil capillary tip test tubes). Undissolved water shall be determined to nearest 10 p/m. The water in oil shall be as specified in 4.6.8.1 (b).

(c) Gravimetric analysis (solids content) - Gravimetric analysis for solid contaminant shall be in accordance with ASTM D 2276. Analysis shall be performed on entire sample. The solids contaminate injection shall be as specified in 4.6.8.1 (c).

4.6.8.3.2 Effluent test methods. Effluent quality test methods shall be as follows:

(a) Sampling procedures. The sample cock shall be fully opened and 200 mL shall be drawn. The 200 mL shall be discarded. Samples shall then be obtained as follows:

(1) Undissolved (free) water and gravimetric analysis - A sample of at least 800 mL shall then be collected for analysis, without closing the sample valve. At least 500 mL shall be used for gravimetric analysis and at least 300 mL used for determination of undissolved water.
(2) Water rejection (oil in water). At least one sample shall be collected in accordance with ASTM D 3370. The sample shall be prepared in accordance with ASTM D 3921.

(b) Gravimetric analysis (solids removal) - Gravimetric analysis for solid contaminant shall be in accordance with ASTM D 2276.

(c) Undissolved (free) water - Undissolved water shall be determined in accordance with ASTM D 3240. Undissolved water shall be determined to the nearest 1 p/m.

(d) Water rejection (oil in water) - The oil-in-water analysis shall be determined in accordance with ASTM D 3921.

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4.6.9 Surge load test. The sample unit shall demonstrate the capability to process influent with water at 200 gal/min without discharging any water from the clean oil outlet. One 5-minute surge test shall be performed on the sample unit at every 300 hours of operation for 1800 hours.

4.6.10 Solids retention test. The ability of the unit to remove solids and water from fuel shall not decline as the solids retention capacity is approached. A solids retention test shall be performed during which effluent samples shall be taken at regular intervals as the purifier is run from a clean bowl condition to the point when the calculated solids retention capacity is approached. These samples will be analyzed as specified in 4.6.8.3. The analyses shall verify that all the samples conform to the requirements of 4.6.8.2.

4.6.11 Vibration test. The sample unit shall be subjected to and meet the type I and II vibration requirements of MIL-STD-167-1.

4.6.12 Shock tests. The sample unit chosen for first article testing shall be subjected to the high impact grade A, class II, type A shock test in accordance with MIL-S-901.

4.6.12.1 Shock test samples. Equipment previously accepted shall not be resubmitted for shock test except when evidence of low shock resistance develops in the units installed. The Command or agency concerned will initiate action with the manufacturer for correction of deficiencies, and may require shock tests at any time.

4.6.13 Drivers. Unless otherwise approved by the Command or agency concerned, the test purifiers shall be shock tested with drivers (see 6.12). Purifiers shock tested with one driver shall not be required to be shock tested again when supplied with a different shock qualified driver. Drivers shall be subject to shock tests in accordance with the applicable equipment specifications.

4.6.13.1 Failure. A purifier unit shall be considered to have failed the shock test in the event of any of the following:

(a) Breakage of any part, including mounting bolts.
(b) Distortion or dislocation of any part, such as shaft, mounting feet, bearings (except for anti-friction bearing brinelling), or other parts, which would render the unit unusable in any emergency.
(c) A mechanical imbalance of more than 2 times the amplitude of balance measured prior to test at rated speed.
(d) The purifier cannot be restarted, or unable to exhibit acceptable performance as specified in 4.6.5.

4.6.14 Electromagnetic interference. To determine conformance to 3.5.6, the purifier shall be qualified in accordance with applicable provisions of MIL-STD-462.

4.7 Individual components. Individual components, such as pumps, controllers, heaters, and so forth, shall be examined and tested at the place of manufacture in accordance with the requirements of the applicable component specification.

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4.8 Inspection of packaging. Sample packages and packs, and the inspection of the preservation, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging or preparation for delivery requirements of referenced documents listed in section 2, see 6.7.)

5.1 Packaging requirements. The packaging (preservation, packing and marking) requirements of purifiers shall be in accordance with MIL-P-16789 for the level (A, C, or commercial) of preservation; level of packing (A, B, C, or commercial), and marking, including packaging acquisitioning options therein as specified (see 6.2). In addition, the following applies:

(a) Navy fire-retardant requirements.

(1) Treated lumber and plywood. When specified (see 6.2), all lumber and plywood, including laminated veneer material used in shipping container and pallet construction, members, blocking, bracing, and reinforcing shall be fire-retardant treated material conforming to MIL-L-19140 as follows:

Levels A and B - Type II - weather resistant.
Category 1 - general use.

Level C - Type I - non-weather resistant.
Category 1 - general use.

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(2) Fiberboard. Fiberboard used in the construction of class domestic, non-weather resistant fiberboard and cleated fiberboard boxes including interior packing forms shall meet the flamespread index and the specific optic density requirements of PPP-F-320.

5.1.1 Marking. In addition to the marking required under 5.1, purifiers shall be marked as critical close tolerance equipment. Unit packs and shipping containers and unpacked shipments shall be marked with the following:

CRITICAL, CLOSE TOLERANCE
OPERATING EQUIPMENT
HANDLE WITH CARE
DO NOT DROP OR SUBJECT
TO SHOCKS OR JARS

Markings shall be stencilled, red color, and applied on two sides and both ends of the container or shipment, letters shall be not less than 1-1/2 inches high, and shall be of such size as to be legible. In addition, arrows and the word "UP", center of balance, sling or lifting point markings shall apply.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Purifiers covered by this specification are intended for use on board Naval surface ships for the removal of water and solids from jet fuel.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number and date of this specification.
(b) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(c) When first article is required (see 3.1).
(d) When resilient mounts are required (see 3.3.2 and 3.3.2.1).
(e) Motors (see 3.3.21).

(1) Type (ac, dc).
(2) Voltage.
(3) Low noise application.

(f) Controllers (see 3.3.22).

(1) Voltage.
(2) Abnormal vibration frequency.
(3) Low noise application.

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(g) When purifier capacity is other than as specified (see 3.5.1).

(1) Size.
(2) Operating fluid.
(3) Capacity.

(h) MIL-STD-167-1 type I frequency range (see 3.5.5.1).
(i) Special tools (see 3.10).
(j) Level of preservation and packing and other packaging acquisition options required (see 5.1).
(k) When fire-retardant material is required (see 5.1(a)).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Description (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
3.2.3	DI-MISC-80678	Certification/data report	10.3.1 does not apply
3.3 and appendix A	DI-DRPR-80651	Engineering drawings	Level 1
3.3.24	DI-V-7002	Provisioning parts list	---
3.3.24	DI-V-7016	Provisioning and other procurement screening data	---
3.3.24	UDI-S-22694	Certificate of prior submission, instrument calibration procedures	---
3.7.1	DI-R-7085	Failure mode, effects, and criticality analysis report	---
3.7.1	DI-RELI-80250	Reliability test plan	---
3.7.1	DI-RELI-80251	Reliability test procedures	---
3.7.2	DI-MNTY-80831	Maintainability/test-ability demonstration test plan	---
3.9	DI-SAFT-80100	System safety program plan	Contractor format
3.9	DI-SAFT-80101	System safety hazard analysis report	Contractor format
4.1.1	DI-QCIC-81110	Inspection and test plan	---
4.6	DI-NDTI-80809	Test/inspection reports	---
4.6	DI-T-23731	Notification of tests	---
4.6	UDI-T-23732	Procedures, test	
4.6.5	DI-RELI-80252	Reliability test reports	---
4.6.6	DI-MNTY-80832	Maintainability/testability demonstration test report	---

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The above DID's were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.5 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.6 Bid data. The following information should be submitted with each bid:

(a) Manufacturer's model and type designation of purifier.
(b) Sectional assembly of purifier or reference to previously approved drawings.
(c) A drawing showing overall dimensions of the unit or reference to previously approved drawings.
(d) Horsepower of driving motors.
(e) Output shaft speed of driving motor and of rotating element.
(f) Weight of complete unit.
(g) Manufacturers of major components such as motors and controllers.
(h) Description of controls.

6.7 Sub-contracted material and parts. The packaging or preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.8 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

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When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.9 Special tools. It is desired that one set of special tools as defined in 3.10 and 6.2 be acquired for each ship's compartment in which purifiers are installed. The number of sets required should be established by the acquisition requesting activity.

6.10 Patent notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government acquisition operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied that said drawings; specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

6.11 Definitions. Lot is defined as a collection of units of products from which a sample is to be drawn and inspected to determine conformance with the acceptability criteria. Each lot should consist of units of product of a single type, grade, class, size and composition, manufactured under the same conditions, and at essentially the same time.

6.12 Reconditioning. Purifiers which have been subjected to high-impact shock tests and have successfully passed this test are considered acceptable for such service as the contracting activity concerned may authorize. Such equipment should be returned to the manufacturer for reconditioning as follows:

(a) Minor deformations of mounting feet should be corrected.
(b) Minor deformations affecting alignment of parts or components should be corrected.
(c) All bearings should be replaced.
(d) Each part should be carefully examined by the manufacturer and any part which he considers substandard shall be replaced.
(e) The rotating element should be rebalanced.
(f) The motor should be corrected as required by the applicable motor specifications.
(g) The reconditioned purifier should be tested in accordance with 4.6.2.

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6.13 Subject term (key word) listing.

Bearings
Brakes
Frames
Lubricants

6.14 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 4330-N094)

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APPENDIX A

ENGINEERING DRAWINGS TECHNICAL CONTENT REQUIREMENTS

10. SCOPE

10.1 Scope. This appendix covers the technical requirements that should be included on drawings when required by the contract or order. This appendix is mandatory only when data item description DI-DRPR-80651 is cited on the DD Form 1423.

20. APPLICABLE DOCUMENTS

20.1 Government documents.

20.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

GGG-P-781 - Puller, Mechanical Puller Attachment, Mechanical, and Puller Set, Mechanical.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, BLDG. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

30. DRAWINGS

30.1 Drawings. When required by the contract or order, drawings shall contain the following information:

(a) A sectional assembly drawing. Every part in the assembly shall be shown. Side or partial views may be used to show parts not otherwise shown in the main section view. All clearances, tightening torques, and thread lubrication requirements shall be shown on the assembly drawing.

(b) Complete parts list. The parts list shall include an indication of each part required to be furnished as an on board repair part (this is not a repair parts list) and include the manufacturer's drawing number and service part number for all parts for which detail drawings are required. Parts shall be named to indicate the function they serve.

(c) Flow diagram and interface drawing showing power, pressure, temperature and flow requirements (as applicable) for all connections to ship's systems.

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(d) Controller drawing showing all indicators, alarms, and controls. It shall include a logic diagram and a parts list.
(e) Special tools shall be detailed and included in the list of materials. Standard complete or limited sets of pullers or parts of puller sets in accordance with GGG-P-781 shall be identified as to type and use in notes on the tool drawings.

30.1.1 Outline drawings. Outline drawings and certification data shall be furnished as supplemental drawings to all sectional assembly and parts list drawings. Outline drawings shall contain the following:

(a) Dimensional outline assembly drawing of the purifier with its motor, bedplate, and attached auxiliaries.
(b) Table of weights of individual components and weight of complete unit (dry and in the operating condition). This shall include the weight of the subbase, fluids, piping, and any other weight that may contribute to loading of the mountings.
(c) Shipbuilder's connections showing size, type, and dimensions of flanges.
(d) Radii of gyration of complete assembly about each of the three principal axes (required when equipment is sound isolated).
(e) Identification of high-impact shock tests, nondestructive tests, and other tests, if performed, and the Command or agency letters approving these reports or extension of any tests.
(f) In addition, the outline drawing shall include overall dimensions of the complete unit including locating dimensions of all fittings and connections, the space required for removal and replacement of parts for maintenance, and mounting information.
(g) Lifting instructions.

30.1.2 "Manufacturer's Use Only" notes. Information intended for manufacturer's use only shall be so designated.

30.1.3 Preferred material reference. Where materials of identical or equal quality can be identified by more than one specification or standard, the drawings shall reference only one such specification or standard. In selecting the specification or standard to be referenced, the following is the order of precedence:

(a) Federal specification
(b) Military specification
(c) Industry and technical society specification or standard
(d) Manufacturer's specification or standard.

Where materials other than covered by (a), (b), and (c) of 30.1.3 are referenced, the drawings shall show the complete chemical and physical properties of the proposed material. In addition, the drawings shall identify the material in terms of the nearest Federal or Military specification, in order to enable Naval repair facilities to make emergency repairs as necessary from Navy material stocks.

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